

Perfect — this is exactly what you needed.

Below is the **updated, clean, complete, single-source-of-truth document** for your **Quiz & Assessment Module**, taking into account your actual implementation style and everything we've discussed.

Save this. Build against this. Refer to this whenever you feel confused.



Quiz & Assessment Module — Requirements & Design Document



Purpose

The Quiz & Assessment module is responsible for:

- Assessing student learning
- Enforcing fair evaluation
- Maintaining data integrity
- Providing meaningful feedback

This module operates **per lesson**.



Core Entities

1. Quiz

- Belongs to **Lesson**
- Has:
 - `total_marks`
 - Status: draft / active
 - Created & updated timestamps

2. Question

- Belongs to **Quiz**
- Has:

- text
- marks
- order
- multiple options

3. Option

- Belongs to **Question**
- Exactly **one correct option**

4. QuizAttempt

- Belongs to **Student**
- Belongs to **Quiz**
- Stores:
 - score
 - submitted_at
 - status (in_progress / submitted)

5. Answer

- Belongs to **QuizAttempt**
- Belongs to **Question**
- Stores chosen option

Pages & Responsibilities

Page	Who Uses It	Purpose
Quiz Management	Instructor	Create & manage questions
Quiz Overview	Student	View quiz info & start attempt
Quiz Attempt	Student	Take quiz
Quiz Result	Student	View score

Permissions & Access Control

Instructor Permissions

Action	Allowed
Create/edit quiz	✓
Add/edit/delete questions	✓
View student results	✓
Attempt quiz	✗

Checks:

- Must own the parent course
- Admin override allowed

Student Permissions

Action	Allowed
View quiz overview	✓
Start quiz attempt	✓
Submit quiz	✓
View result	✓
Edit quiz/questions	✗

Checks:

- Must be enrolled
- Course must be published

Admin Permissions

- Full override on everything

Validation & Security Rules

Quiz Visibility (Quiz Overview Page)

A student may view the quiz overview if:

1. User is authenticated
 2. Course is published
 3. Student is enrolled
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Starting a Quiz Attempt

A student may start if:

1. User is authenticated
 2. Role is **student**
 3. Course is published
 4. Student is enrolled
 5. Lesson is completed (if sequential learning enforced)
 6. Student has **no previous submitted attempt**
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Attempting the Quiz

- Attempt belongs only to the student
 - Cannot access others' attempts
 - Cannot modify after submission
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Submitting the Quiz

Allowed if:

1. Attempt belongs to student
2. Attempt not already submitted

On submission:

- Score is calculated automatically
- Attempt is locked
- Result becomes immutable



Scoring Rules

- Each question has its own `marks`
 - Sum of question marks = `quiz.total_marks`
 - Score computed automatically
 - No manual editing of scores by students
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Data Integrity Constraints

Rule	Required
One attempt per student per quiz	✓
Question order unique per quiz	✓
Option count ≥ 2	✓
Exactly one correct option	✓
Question marks > 0	✓



Failure Handling

Condition	Result
Not enrolled	Access denied
Attempt already submitted	Block retry
Instructor tries to attempt	Denied
Tampered request	Reject



Design Principles

A quiz is a **secure transaction**

A result is **immutable**

Assessment must be **fair and verifiable**



Interview Summary Line

“The quiz system enforces strict access control, validates enrollment and attempt ownership, ensures scoring integrity, and prevents tampering through immutable submissions.”

If you'd like, I can now:

- Convert this into a **development checklist**
- Or help you map this onto your current code structure